

June 2026

Geo AI Master 2 – Lebanese University, Faculty of science

Course: AI for earth observation

AI4EO Course Project

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Research Paper Selection

For this project, our team selected the research paper "**Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation**" by Tomoaki Yamaguchi et al. The paper is available on arXiv and is accompanied by a publicly accessible implementation repository hosted on Zenodo.

Paper: *Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation* (2025)

[2512.21066v3.pdf](#)

[\[2512.21066\] Agentic Explainable Artificial Intelligence \(Agentic XAI\) Approach To Explore Better Explanation](#)

Implementation Repository: Zenodo Repository (DOI: 10.5281/zenodo.17876330)
[Supplementary Materials for: Agentic Explainable Artificial Intelligence \(Agentic XAI\) Approach To Explore Better Explanation](#)

We selected this paper because it directly addresses the topic of **Explainable AI (XAI) for Agentic AI systems**, which is the main focus of this course project. The paper introduces a novel concept called **Agentic XAI**, where a large language model (LLM) acts as an autonomous agent that iteratively improves and refines explanations generated from XAI outputs. The proposed framework combines SHAP-based explainability with agentic AI techniques to produce more understandable and actionable explanations for end users.

This paper satisfies all the project requirements:

1. Recent and Relevant Research

The paper is a recent study published in 2025 and focuses on the integration of Explainable AI and Agentic AI. It investigates how autonomous AI agents can iteratively refine explanations and improve the communication of machine learning results, making it highly relevant to the course topics.

2. Publicly Accessible Implementation

The authors provide a complete implementation repository through Zenodo. The repository includes prompts, Python code, generated visualizations, recommendations, and all materials required to reproduce the Agentic XAI workflow.

3. Sufficient Experimental Details for Reproducibility

The paper presents a detailed methodology, including the dataset description, machine learning model configuration, SHAP analysis process, iterative refinement workflow, evaluation metrics, and statistical analysis procedures. Furthermore, the repository contains the code and outputs for every refinement round, which strongly supports reproducibility of the reported results.

Overall, this paper is an excellent candidate for the project because it combines explainability, large language models, and agentic reasoning in a practical framework, while also providing open-source resources and detailed experimental documentation that facilitate reproduction and further analysis.